

Oil Well Drilling

Labwork 1: IIA4117

This is a model of an oil well drilling with a single function of turning on or off the mud pump. A general drawing of such a rig can be seen in figure 1.

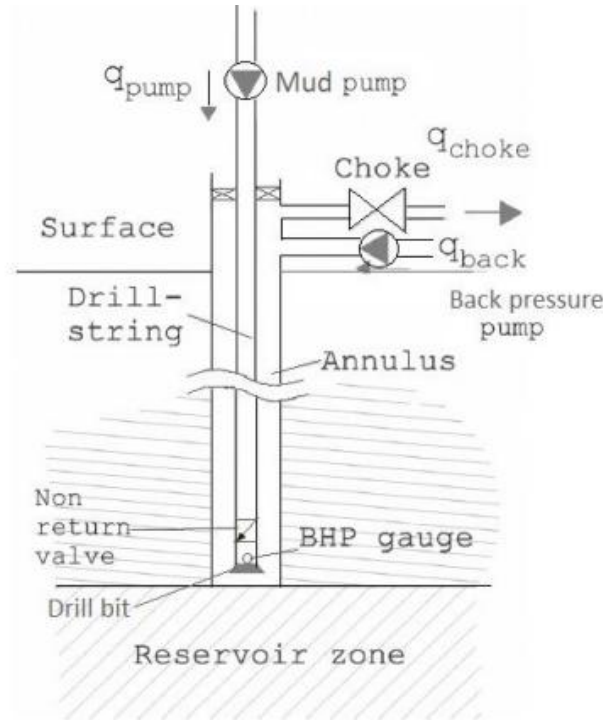


Figure 1. Simplified diagram of an oil well drilling system. [1]

The model for the system can be seen in the equations below.

$$\frac{dP_p}{dt} = \frac{\beta_d}{A_d L} (q_{pump} - q_{bit})$$

$$\frac{dq_{bit}}{dt} = \begin{cases} \frac{A_d}{\rho_l L} (P_p + \rho_l g L - \Delta P_f^d - P_{bit}) & q_{bit} > 0 \\ \max \left\{ 0, \frac{A_d}{\rho_l L} (P_p + \rho_l g L - \Delta P_f^d - P_{bit}) \right\} & q_{bit} = 0 \end{cases}$$

$$\frac{dP_c}{dt} = \frac{\beta_a}{A_a L} (q_{bit} + q_{res} + q_{back} - q_{choke})$$

$$\frac{dL}{dt} = v_{pen}$$

$$P_{bit} = P_c + \rho_{mix} g L + \Delta P_f^a$$

$$\rho_{mix} = \rho_w W_c + (1 - W_c) \rho_l$$

$$q_{res} = \max \{ 0, PI(P_{res} - P_{bit}) \}$$

$$q_{choke} = \frac{27.3}{3600 \sqrt{10^5}} Z_c(u_c) \sqrt{\frac{\max\{0, P_c - P_0\}}{\rho_{mix}}}$$

$$Z_c(u_c) = \begin{cases} 0 & u_c < 5 \\ 0.111u_c - 0.556 & 5 \leq u_c < 50 \\ 0.5u_c - 20 & 50 \leq u_c \end{cases}$$

$$\Delta P_f^x = \frac{f_d L \rho_x v_x^2}{2 D_h^x} \quad x = d \vee a$$

To simulate this, values from table 1 were used

Table1: Values for the parameters and variables

Parameters / Variables	Value	Unit	Description
ρ_l	1150	kg/m^3	Drill mud density
ρ_w	1000	kg/m^3	Water density
WC	0.1		Water cut of the reservoir fluid
A_d	0.0067	m^2	Cross sectional area of drill string
A_a	0.278	m^2	Cross sectional area of annulus
D_d	0.0925	m	Hydraulic diameter of drill string
D_a	0.211	m	Hydraulic diameter of annulus
L	1600	m	Vertical depth of the well
PI	1.6667×10^{-9}	m^5/Ns	Productivity Index value
P_{res}	250×10^5	N/m^2	Reservoir pressure
P_{frac}	270×10^5	N/m^2	Fracture pressure
P_{coll}	220×10^5	N/m^2	Collapse pressure
P_{bit}^{ref}		N/m^2	Reference pressure
P_0	4×10^5	N/m^2	Pressure downstream the choke valve
ϵ/D_d	10^{-5}		Relative roughness of pipe in drill string
ϵ/D_a	10^{-4}		Relative roughness of pipe in the annulus
μ_d	0.015	kg/ms	Dynamic viscosity of the drill fluid
β_d	3×10^8	N/m^2	Bulk modulus in the drill string
β_a	2.4×10^8	N/m^2	Bulk modulus in the annulus
q_{pump}^{nom}	1500	l/min	Nominal flow rate of the drill fluid
u_c^{nom}	70	%	Nominal choke valve opening
g	9.81	m/s^2	Acceleration due to gravity
f_d	0.02		Friction factor

The following initial conditions were also used:

$$P_p = 5 \cdot 10^6 [Pa]$$

$$q_{bit} = 0.025 [m^3/s]$$

$$P_c = 5.5 \cdot 10^6 [Pa]$$

$$q_{back} = 0 [m^3/s]$$

$$\Delta q_{pump} = \pm 300 [l/min^2]$$

The Simulink setup created can be seen in figure 2

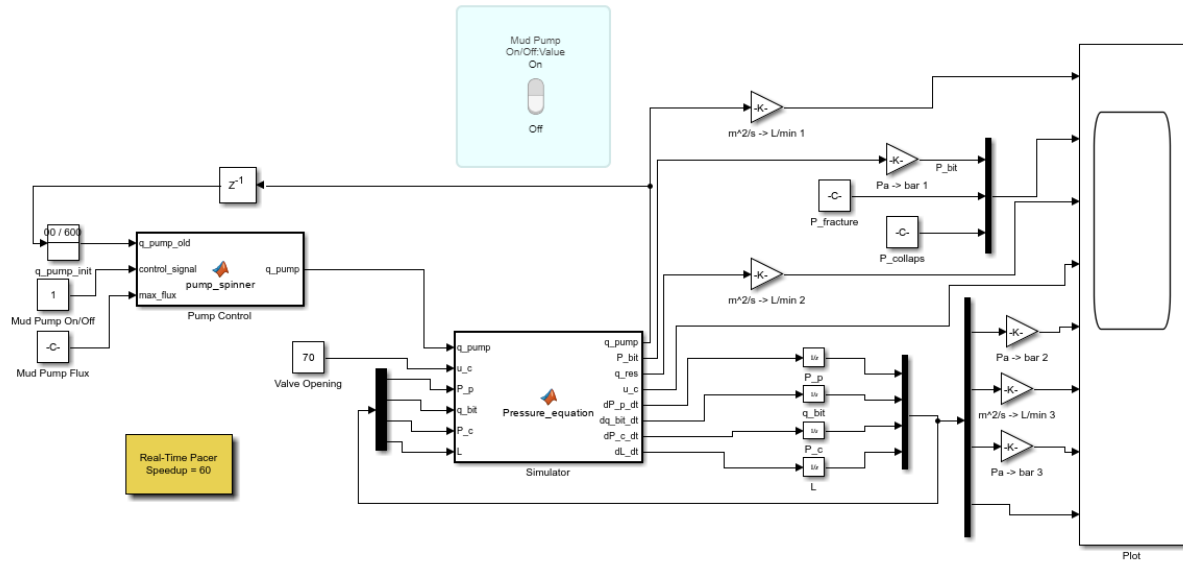


Figure 2: Simulink setup for modeling.

In figure 3 and figure 4, the pump_spinner and Pressure_equations MATLAB functions can be seen.

```
function q_pump = pump_spinner(q_pump_old, control_signal, max_flux)
    step = 30/60000 / 60;
    q_pump = q_pump_old;

    if control_signal > 0
        if q_pump_old < max_flux
            q_pump = q_pump_old + step;
        end
        if q_pump > max_flux
            q_pump = max_flux;
        end
    elseif control_signal <= 0
        if q_pump_old > 0
            q_pump = q_pump_old - step;
        end
        if q_pump < 0
            q_pump = 0.0;
        end
    end
end
```

Figure 3: pump_spinner MATLAB function.

```

function [q_pump, P_bit, q_res, u_c, dP_p_dt, dq_bit_dt, dP_c_dt, dL_dt] = Pressure_equation(q_pump, u_c, P_p, q_bit, P_c, L)
% Initial conditions
% P_p                                # Mud pump downstream pressure
% q_bit                              # Volum flux of drill mud through drill bit
% P_c                                # Well head pressure at annulus
% L = (1600) m                       # Depth of well

% Constants
% u_c = (70) \%                      # Choke Valve opening
% q_pump (0.025) m^3/s                # Volum flux of drill mud through mud pump
beta_d = 3 * 10^8; % N/m^2            # Bulk modulus at drill string
A_d = 0.0067; % m^2                  # Area of drill string cross section
g = 9.81; % m/s^2                    # Gravitational acceleration
beta_a = 2.4 * 10^8; % N/m^2         # Bulk modulus at annulus
A_a = 0.278; % m^2                   # Area of annulus cross section
q_back = 0; % m^2/s                  # Volum flux of backpressure pump
P_0 = 4 * 10^5; % N/m^2              # Pressure downstream from choke valve
rho_w = 1000; % kg/m^3               # Water density
W_c = 0.1; %                         # Water cut/ratio in reservoir fluid
rho_l = 1150; % kg/m^3               # Density of drill mud
P_i = 1.6667 * 10^(-9); % m^5/Ns     # Productivity index of well
P_res = 2.5 * 10^7; % N/m^2          # Reservoir pressure
v_pen = 0; % m/s                     # Drilling speed
D_hd = 0.0925; % m                   # Hydraulic diameter of drill string
D_ha = 0.211; % m                    # Hydraulic diameter of annual

% Parameters
rho_mix = rho_w * W_c + (1 - W_c) * rho_l;

%Z_c = max([0, 0.111 * u_c - 0.556, 0.5 * u_c - 20]);
Z_c = calc_Z_c;

q_choke = 27.3 / (3600 * sqrt(10^5)) * Z_c * sqrt(max([0, P_c - P_0]) / rho_mix);

DeltaP_fd = calc_DeltaP_f(rho_l, q_pump, A_d, D_hd);
DeltaP_fa = calc_DeltaP_f(rho_mix, q_choke, A_a, D_ha);

P_bit = P_c + rho_mix * g * L + DeltaP_fa;

q_res = max([0, P_i * (P_res - P_bit)]);

% Differnetial equations
dP_p_dt = beta_d / (A_d * L) * (q_pump - q_bit);

if q_bit <= 0
    dq_bit_dt = max([0, A_d / (rho_l * L) * (P_p + rho_l * g * L - DeltaP_fd - P_bit)]);
else
    dq_bit_dt = A_d / (rho_l * L) * (P_p + rho_l * g * L - DeltaP_fd - P_bit);
end

dP_c_dt = beta_a / (A_a * L) * (q_bit + q_res + q_back - q_choke);

dL_dt = v_pen;

% Functions
function res = calc_Z_c
    if 0.111 * u_c - 0.556 < 0
        res = 0;
    elseif u_c < 50
        res = 0.111 * u_c - 0.556;
    else
        res = 0.5 * u_c - 20;
    end
end

function res = calc_DeltaP_f(rho_x, q_x, A_x, D_hx)
    f_d = 0.02; % # Friction factor.

    v_x = q_x / A_x;
    res = f_d * L * rho_x * v_x^2 / (2 * D_hx);
end
end

```

Figure 4: Pressure_equations MATLAB function.

Running this simulation, for 75 minutes, gives the plot seen in figure 5, where two breaks in the pumping of mud have been added manually via the button to control the mud pump.

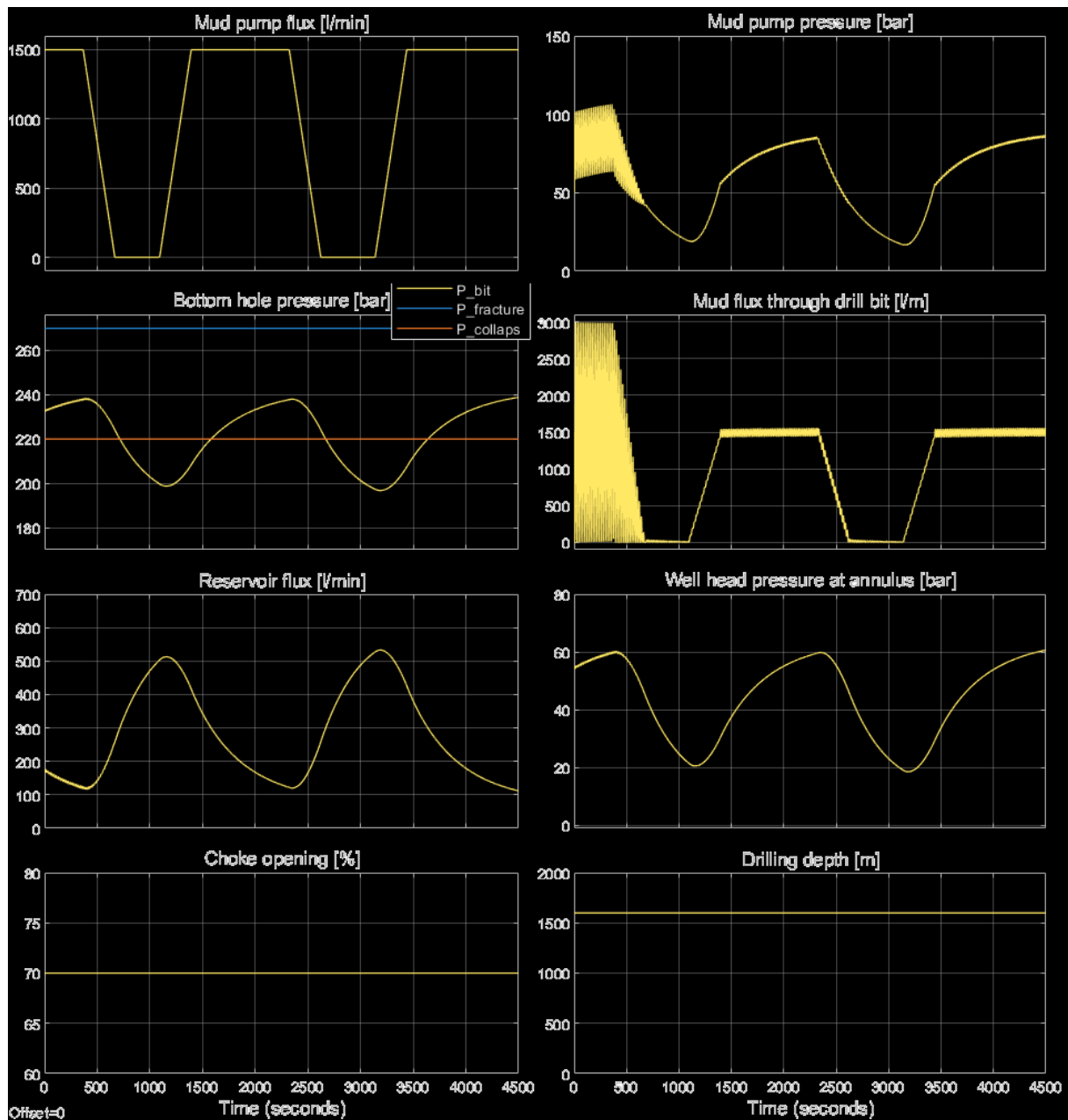


Figure 5: Results from running the simulation.

The results on the left-hand plots are basically the same as the once found in figure 3 of [1]. This is a sign that the simulation is done correctly. Another sign is that pressure sinks when the pump is turned off and rises when the pump is turned on. This follows naturally from the fact that more stuff pumped into the well means higher pressure and vice versa.

From the right-hand side plots in figure 5, mud pump pressure and mud flux through drill bit are quite unstable in the beginning. They are in fact very unstable until the mud pump

is turned off once, then become much more stable after that. If the mud pump is never turned off, they are unstable for the whole simulation. This is likely due to some instability with the initial condition, but it does however not seem to have any effect on the rest of the model. This is likely due to very high oscillation frequency.

Reference

- [1] R. Sharma, "Model Predictive Control," 25 August 2022. [Online]. Available: <https://web01.usn.no/~roshans/mpc/downloads/oil-well-drilling-labwork1.pdf>.